

Scheduling Algorithms

Mohammad Rehan

2520030066

1. FCFS Scheduling algorithm

```
rehan_lenovo@Rehan-Lenovo × + v
GNU nano 8.7.1
#include <stdio.h>

int main() {
    int n, i;
    int bt[20], wt[20], tat[20];

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time for each process:\n");

    for (i = 0; i < n; i++) {
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);
    }
    wt[0] = 0;

    for (i = 1; i < n; i++) {
        wt[i] = wt[i - 1] + bt[i - 1];
    }
    for (i = 0; i < n; i++) {
        tat[i] = wt[i] + bt[i];
    }

    printf("\nProcess\tBT\tWT\tTAT\n");

    for (i = 0; i < n; i++) {
        printf("P%d\t%d\t%d\t%d\n",
            i + 1, bt[i], wt[i], tat[i]);
    }

    return 0;
}

^G Help      ^O Write Out  ^F Where Is   ^K Cut
^X Exit      ^R Read File  ^\ Replace    ^U Paste
```

Output:

```
rehan_lenovo@Rehan-Lenovo × + v
rehan_lenovo@Rehan-Lenovo:~/OS$ nano FCFS.c
rehan_lenovo@Rehan-Lenovo:~/OS$ nano FCFS.c
rehan_lenovo@Rehan-Lenovo:~/OS$ gcc FCFS.c -o FCFS
rehan_lenovo@Rehan-Lenovo:~/OS$ ./FCFS
Enter number of processes: 4
Enter burst time for each process:
P1: 5
P2: 3
P3: 8
P4: 6

Process BT      WT      TAT
P1      5       0       5
P2      3       5       8
P3      8       8      16
P4      6      16      22
rehan_lenovo@Rehan-Lenovo:~/OS$ |
```

2. SJF_NONpreemptive

```
#include <stdio.h>

int main() {
    int n, i, j;
    int bt[20], wt[20], tat[20];
    int temp;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time:\n");

    for (i = 0; i < n; i++) {
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);
    }

    // Sort burst times
    for (i = 0; i < n - 1; i++) {
        for (j = i + 1; j < n; j++) {
            if (bt[i] > bt[j]) {
                temp = bt[i];
                bt[i] = bt[j];
                bt[j] = temp;
            }
        }
    }

    // Waiting time
    wt[0] = 0;

    for (i = 1; i < n; i++) {
        wt[i] = wt[i - 1] + bt[i - 1];
    }

    // Turnaround time
    for (i = 0; i < n; i++) {
        tat[i] = wt[i] + bt[i];
    }

    printf("\nProcess\tBT\tWT\tTAT\n");

    for (i = 0; i < n; i++) {
        printf("P%d\t%d\t%d\t%d\n",
            i + 1, bt[i], wt[i], tat[i]);
    }

    return 0;
}
```

Output:

```
rehan_lenovo@Rehan-Lenovo:~/OS$ nano SJF_np.c
rehan_lenovo@Rehan-Lenovo:~/OS$ gcc SJF_np.c -o SJF_np
rehan_lenovo@Rehan-Lenovo:~/OS$ ./SJF_np
Enter number of processes: 5
Enter burst time:
P1: 7
P2: 3
P3: 10
P4: 2
P5: 5

Process BT      WT      TAT
P1      2        0        2
P2      3        2        5
P3      5        5       10
P4      7       10       17
P5     10       17       27
rehan_lenovo@Rehan-Lenovo:~/OS$ |
```

3. Priority

```
GNU nano 8.7.1
#include <stdio.h>

int main() {
    int n, i, j;
    int bt[20], priority[20], wt[20], tat[20];
    int temp;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time and priority:\n");

    for (i = 0; i < n; i++) {
        printf("P%d Burst Time: ", i + 1);
        scanf("%d", &bt[i]);

        printf("P%d Priority: ", i + 1);
        scanf("%d", &priority[i]);
    }

    for (i = 0; i < n - 1; i++) {
        for (j = i + 1; j < n; j++) {
            if (priority[i] > priority[j]) {
                temp = priority[i];
                priority[i] = priority[j];
                priority[j] = temp;

                temp = bt[i];
                bt[i] = bt[j];
                bt[j] = temp;
            }
        }
    }
}
```

Output:

```
rehan_lenovo@Rehan-Lenovo × + v
rehan_lenovo@Rehan-Lenovo:~/OS$ nano Priority.c
rehan_lenovo@Rehan-Lenovo:~/OS$ gcc Priority.c -o Priority
rehan_lenovo@Rehan-Lenovo:~/OS$ ./Priority
Enter number of processes: 5
Enter burst time and priority:
P1 Burst Time: 7
P1 Priority: 3
P2 Burst Time: 4
P2 Priority: 1
P3 Burst Time: 6
P3 Priority: 4
P4 Burst Time: 2
P4 Priority: 2
P5 Burst Time: 5
P5 Priority: 5

Process BT      Priority      WT      TAT
P1      4      1      0      4
P2      2      2      4      6
P3      7      3      6      13
P4      6      4      13      19
P5      5      5      19      24
rehan_lenovo@Rehan-Lenovo:~/OS$
```

4. ROUND ROBIN

```
rehan_lenovo@Rehan-Lenovo × + v
GNU nano 8.7.1 RoundRobin.c *
#include <stdio.h>

int main() {
    int n, i;
    int bt[20], rem_bt[20];
    int wt[20], tat[20];
    int quantum;
    int time = 0;
    int done;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time:\n");

    for (i = 0; i < n; i++) {
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);

        rem_bt[i] = bt[i];
        wt[i] = 0;
    }

    printf("Enter time quantum: ");
    scanf("%d", &quantum);

    while (1) {
        done = 1;
```

Output:

```
rehan_lenovo@Rehan-Lenovo X + v
rehan_lenovo@Rehan-Lenovo:~/OS$ nano RoundRobin.c
rehan_lenovo@Rehan-Lenovo:~/OS$ gcc RoundRobin.c -o RoundRobin
rehan_lenovo@Rehan-Lenovo:~/OS$ ./RoundRobin
Enter number of processes: 4
Enter burst time:
P1: 5
P2: 4
P3: 2
P4: 7
Enter time quantum: 2

Process BT      WT      TAT
P1      5      10      15
P2      4       8      12
P3      2       4       6
P4      7      11      18
rehan_lenovo@Rehan-Lenovo:~/OS$ |
```